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* What is Zeek?

- Zeek (formerly Bro) is a passive network analysis framework. Unlike Wireshark which shows you raw packets, Zeek automatically generates structured log files from network traffic — making it much faster to analyse large captures.

PCAP file / Live traffic → Zeek

↓

Structured log files (.log)
conn.log | dns.log | http.log | ssl.log | files.log

* Zeek vs Wireshark vs NetworkMiner

Feature	Wireshark	Network Miner	Zeek
Output	Raw packets	Auto - extracted files	Structured logs
Speed	Slow (manual)	Fast (auto)	Very fast (scripted)
Best for	Deep inspection	Rapid triage	Large scale analysis
Scripting	No	No	Yes (Zeek Scripts)
Used by	Everyone	SOC analysts	Enterprise SOC
Log form	.pcap	GUI tabs	.log

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* Zeek Log Files — The most important Ones

1) conn.log — Every Connection :

The most important zeek log. Record every network connection.

ts → Timestamp

id.orig_h → Source IP

id.orig_p → Source Port

id.resp_h → Destination IP

id.resp_p → Destination Port

proto → Protocol (tcp/udp)

duration → How long connection lasted

orig_bytes → Bytes sent by ~~destination~~ source.

resp_bytes → Bytes sent by destination

conn_state → Connection state

- DFIR use : Long duration connections = possible C2 beaconing. Large resp_bytes = possible data download. Large orig_bytes = possible exfiltration

2) dns.log — Every DNS query :

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Records every DNS query and response.

ts → Timestamp

id.orig_h → Who asked

query → Domain queried

qtype_name → Query type (A, AAAA, MX)

answers → IP address returned

rcode_name → NOERROR or NXDOMAIN

- OFIR USE: rcode_name == NXDOMAIN = DGA malware failing. Repeated queries to same domain = beaconing.

3) http.log — Every HTTP Request:

Records every HTTP request and response.

ts → Timestamp

id.org_h → Client IP

host → Website visited

uri → Path requested

method → GET / POST

user_agent → Browser / client used

status_code → 200, 404, etc

resp_body_len → Response size

- DFIR use: Unusual user_agent = malware. POST method = possible exfiltration. /gate.php, /panel/ in URI = C2

4) ssl.log — Every TLS Connection:

Records TLS / SSL handshake information

server_name → SNI — destination domain

issuer → Certificate issuer

validation_status → Certificate valid?

version → TLS version

- DFIR use: server_name reveals C2 domain even in encrypted traffic. Self-signed certificates = suspicious

5) files.log — Every File Transferred

Records every file seen in network traffic.

filename → File name

mime_type → File type

md5 → File hash

sha256 → File hash

source → Protocol (HTTP, SMTP)

- DFIR use: MD5/SHA256 hash can be checked on VirusTotal to identify malware.

* Zeek-cut → The key Command

zeek-cut is used to select specific columns from zeek logs:

```
# Get source and destination IPs from conn.log
cat conn.log | zeek-cut id.orig_h id.resp_h

# Get DNS queries
cat dns.log | zeek-cut query rcode_name

# Get HTTP URIs and user agents
cat http.log | zeek-cut host uri user_agent method

# Find long connections (possible beaconing)
cat conn.log | zeek-cut id.orig_h id.resp_h duration | sort -k3 -rn

# Find NXDOMAIN responses (DGA)
cat dns.log | zeek-cut query rcode_name
```

| grep NXDOMAIN

Find POST requests
cat http.log | zeek-cut method host uri | grep POST

* Connection States in conn.log

State	Meaning	DFIR Note
SF	Normal established + closed	Normal
S0	SYN sent, no response	Port scan
S1	Established, not closed	Possible C2
REJ	Connection rejected	Port scan / firewall
RSTO	Reset by originator	Suspicious
OTH	No SYN seen	Partial capture